Spatial Analysis of Chlorophyll Concentration

Spencer Matthews - STATS 295 - Winter 2022

Motivation

Marine biologists are often concerned with the amount of marine phytoplankton (a type of plankton that uses photosynthesis to produce energy) in the ocean. These phytoplankton play a pivotal role in the ecosystem of our oceans, and they also help reduce the amount of carbon dioxide in the atmosphere (Wu et al., 2008). There are various factors that affect where phytoplankton can live, and frequently we seek to understand how this system works so that we better understand human impacts on the process (Field et al., 1998).

One of the best proxies for phytoplankton presence is the concentration of Chlorophyll in the ocean water. This chlorophyll can be used to identify regions where phytoplankton are more abundant, which gives target locations for scientific expeditions to study the populations of phytoplankton (Li et al., 2012). In this analysis, we will study the chlorophyll concentration in the Indian Ocean in September of 2021 and analyze it using spatial statistical methods. The goal of this analysis is to understand if there is spatial correlation in where phytoplankton occur that cannot be explained by other predictors. If there is indeed spatial correlation, we will then be able to identify phytoplankton "hot-spots", or locations where the phytoplankton are more likely to occur so that studies can be planned in optimal locations.

Model

In order to analyze the data on chlorophyll concentration we decided to implement a Bayesian model with Conditionally AutoRegressive (CAR) random effects. This approach was chosen due to the form of the data, which came at different granularities. Thus, the easiest way to model it was to create a grid inside the Indian Ocean and model the data as areal data. In the end, we created a regular grid encompassing the Indian Ocean, which led us to having 6,587 area units that were modeled. This model takes the form of Bayesian Hierarchical model shown in Equation 1.

$$\begin{aligned}
&\mathbf{Y}|\beta, \eta', \sigma^2 \sim MVN_N(\mathbf{X}\beta + \eta', \sigma^2\mathbf{I}) \\
&\eta_i'|\eta' j, j \in \partial i \sim N\left(\sum_{j \in \partial i} \frac{w_{ij}}{w_{i+}}\eta_j', \frac{\tau^2}{w_{i+}}\right) \\
&\beta, \tau^2, \sigma^2|\text{Hyperparameters} \sim p(\beta, \tau^2, \sigma^2|\text{Hyperparameters})
\end{aligned} \tag{1}$$

Four predictors of chlorophyll concentration were identified based on knowledge of the factors affecting phytoplankton life: sea salinity, ocean depth, ocean temperature, and the presence of coral. Data was obtained on each of these predictors in the Indian Ocean (during September 2021 for those predictors that are variable). Data on chlorophyll concentration, sea salinity, ocean temperature, and the presence of coral was obtained from MyNASAData.gov (Graham et al., 2022), and data on ocean depth was obtained using the `marmap` package in `R` (Pante and Simon-Bouhet, 2013). In Figure 1 we have choropleth maps of our grid in the Indian Ocean showing the response as well as the linear predictors in our model. As can be seen, both our target outcome and our predictors all seem to exhibit spatial correlation between the areal units. Note that since the concentration was quite skewed, we decided to perform a log transformation and model the log(Chlorophyll Concentration), as is common practice when working with concentrations. Our model for areal unit i is given in Equation 2.

$$\log(\text{Chlorophyll Conc.})_i = \beta_0 + \beta_1(\text{Depth})_i + \beta_2(\text{Salinity})_i + \beta_3(\text{Temp})_i + \beta_4 \mathbb{1}(Coral = 1)_i + \eta_i \tag{2}$$

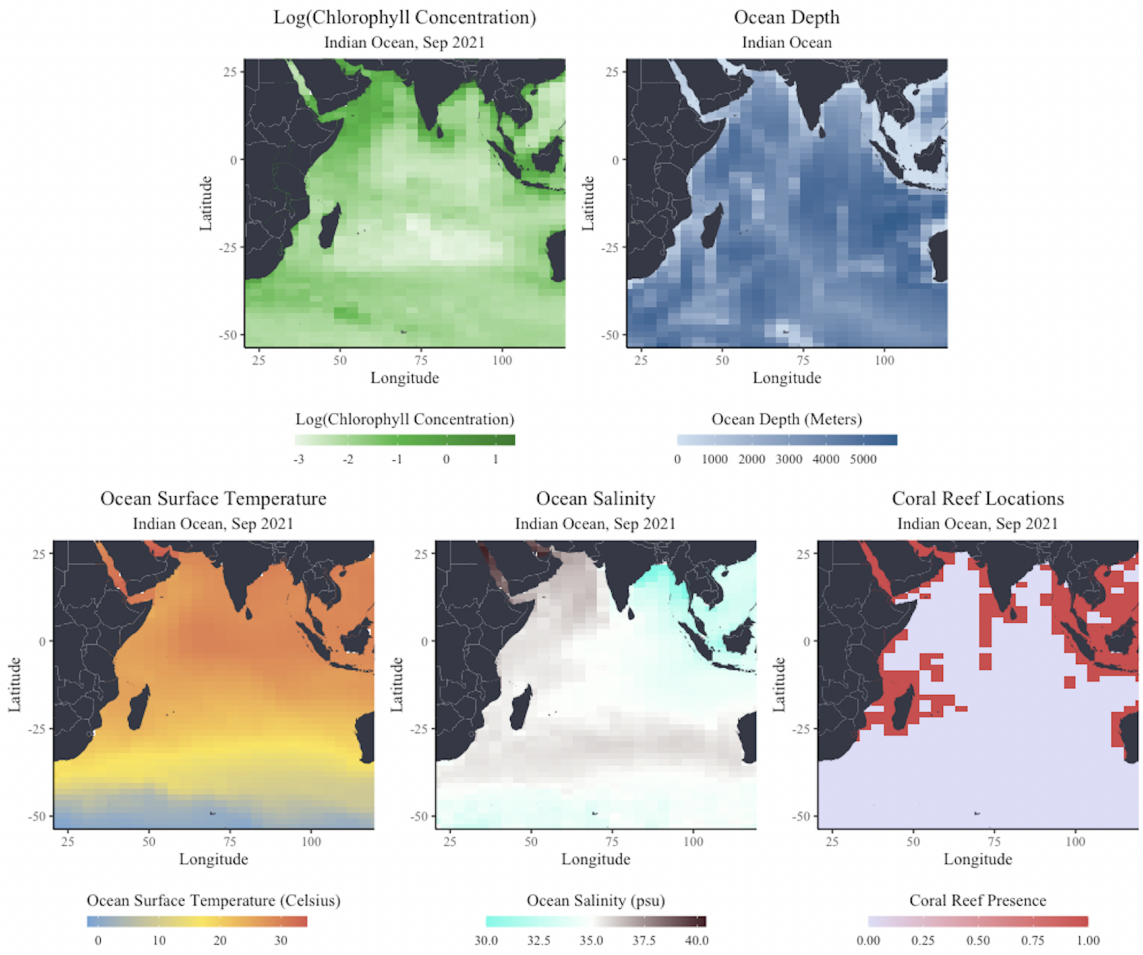

Figure 1: Map of Log(Chlorophyll Concentration) in the Indian Ocean, September 2021 along with the predictors used in our linear model: Depth, Salinity, Temperature, and Coral Presence.

Methods and Inference

Before working on the spatial part of the model, we first fit just the linear model of the log(Chlorophyll Concentration) against our four predictors using OLS to obtain our parameter estimates for β. Estimated values, along with confidence intervals and p-values are given in Table 1.

We see that additional depth is associated with less chlorophyll, while increased salinity, increased temperature, and coral presence are all associated with more chlorophyll. This fits prior hypotheses. Furthermore, most of our predictors are significant, with the exception of coral presence. This may not be significant just because it is highly correlated with depth and temperature, as those are two of the main factors that govern where coral can live.

After fitting the linear model, we were able to pull off the residuals of the model for each areal unit. A weights matrix was computed for our dataset, which resulted in several (about 10) areal units being dropped in order to remove islands. Once the weights matrix and neighbor matrix were computed, we were able to continue our exploratory analysis by computing Moran's I. This was computed both using the asymptotic approximation as well as doing a Monte-Carlo Simulation test. Results of both of these test are given in Table 2.

Table 1: Initial OLS estimates for model parameters without accounting for any spatial correlation in the model.

		95% CI			
	Estimate	Lower Bound	Upper Bound	Test Statistic	P-Value
(Intercept)	-1.5775	-2.0605	-1.0944	-6.4017	0.0000
depth	-0.0003	-0.0003	-0.0003	-45.9337	0.0000
salinity	0.0196	0.0055	0.0337	2.7333	0.0063
temp	0.0025	0.0009	0.0041	3.0450	0.0023
coral	0.0331	-0.0272	0.0934	1.0761	0.2819

Table 2: Results of testing Moran's I, the null hypothesis being that there is no spatial correlation in our residuals. (Alternative hypothesis: greater than)

Description	Value
Moran's I via Asymptotic Test	
Test Statistic	123.98
P-Value	0
Moran's I via Simulation	
Number of Simulations	9,999
Observed Rank	10,000
P-Value	0

We see that the asymptotic test statistic is extremely large resulting in a p-value of essentially zero, and that in our simulation test the observed data was the highest rank resulting in a p-value of essentially zero as well. These small p-values lead to us to conclude that there is indeed spatial correlation between areal units in the residuals, even after accounting for the covariates in our model. This was expected, since plant life tends to clump together and there is a fairly large amount of chance as to where exactly that happens, assuming that conditions are good enough for live to thrive. The extent of this spatial correlation in the residuals can be seen in Figure 2 where positive residuals are shown in green and negative residuals shown in purple. Clearly, there is still spatial correlation in this choropleth map.

Once the exploratory analysis was completed, we fit the model in Equation 1 using an improper CAR model. This was done in `R` using the function `S.CARleroux()` from the `CARBayes` package (Lee, 2013) with a fixed value of 1 for ρ. The β vector had a prior of a normal distribution with a mean of zero and variance 100,000. Both τ^2 and σ^2 had priors that were Gamma(shape $= 1, \text{Scale} = 0.01$). We used the same covariates discussed thus far in the report, and ran the sampler 100,000 iterations. The first 50,000 were used as burn-in and the remaining 50,000 were thinned by selecting every tenth sample. Thus in the end we had 5,000 samples from our posterior distribution for each parameter.

Results

After performing the simulation, we summarized the sampling and posterior distributions for all parameters, which can be seen in Table 3.

Comparing these to the original estimates in Table 1, we see that they are quite different. In our updated model, the intercept term has a credible interval that encompasses zero, but the rest of our linear predictors all have credible intervals that are either entirely greater than zero or entirely less than zero.

Depth still has a negative coefficient, and coral still has a positive one, but both Salinity and Temperature have coefficients that were positive in the initial regression model but ended up being negative in our

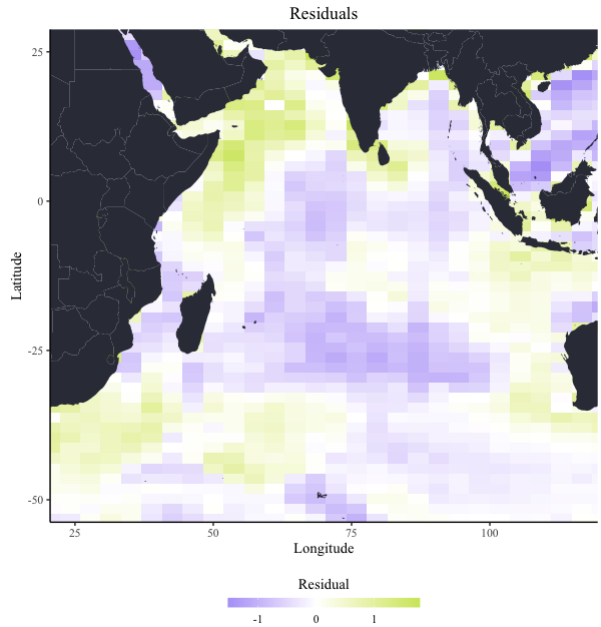

Figure 2: Map of residuals for each areal unit after fitting the OLS model described in the previous section. These residuals were used to test for additional spatial effects as described in this section.

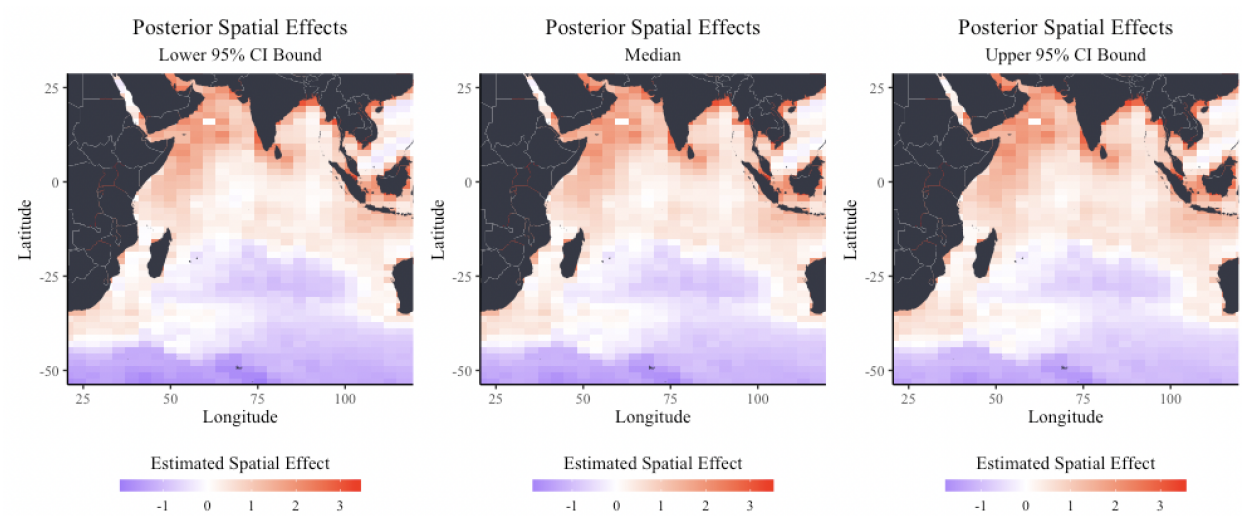

Figure 3: Map of the posterior median Spatial effects for Log(Chlorophyll Concentration), along with the lower and upper bounds of the 95% confidence interval. Due to the number of samples/burn-in required to ensure convergence in our model, these three plots are almost identical.

Table 3: (September 2021 Data) Results and summaries of the posterior distribution for all parameters.

	Median	2.5%	97.5%	Effective Sample Size	Geweke Diagnostic
Intercept	0.4512	-0.0333	1.1070	35.5	0.5
Depth	-0.0001	-0.0002	-0.0001	45.5	1.2
Salinity	-0.0181	-0.0363	-0.0050	37.4	-0.5
Temperature	-0.0678	-0.0713	-0.0633	9.9	-0.1
Coral Presence	0.0821	0.0518	0.1142	165.4	0.8
σ^2	0.0007	0.0005	0.0010	252.0	-0.2
τ^2	0.4673	0.4515	0.4836	5000.0	-0.1
ρ	1.0000	1.0000	1.0000	NA	NA

hierarchical model. Note that the Geweke diagnostic is at an acceptable level for all parameters, and this was confirmed by visual inspection of trace plots for all parameters.

The parameter estimates shown in Table 3 allow us to estimate each η_i from Equation 2. As a part of the sampling process, there were sampled as well and we had 5,000 posterior samples for each areal unit. In Figure 3, we see plots of the posterior distribution of η_i for each areal unit in the Indian Ocean. This plot shows both the lower and upper bounds of a 95% confidence interval, as well as the posterior median. Note that there was not a ton of variation in the η_i's, which means that these three plots are very similar to each other, although there are slight differences.

Looking at Figure 3 we can see that the areal units in red are areas that tend to have more chlorophyll naturally than other areal units, after accounting for the effects of ocean depth, temperature, salinity, and coral presence. These are the areas that would be worth investigating in order to study phytoplankton that produce the chlorophyll. In contrast, those areal units that are more blue/purple are those that are less inclined to have high chlorophyll concentration.

Predictive Power and Future Work

A natural question that arose while working with this model was, "how good is it?" Especially considering that the model was built specifically on data from September of 2021, we naturally wondered what kind of predictive power it had on data from a completely different time frame. To this end, we downloaded the same data as described above for February of 2020 to test the model's ability to predict chlorophyll concentration at the opposite time of year. When we did this and predicted on the February 2021 data set, we saw that the model was fairly robust, with an MSE of 0.7031. Obviously, however, this is much more than the almost non-existent MSE on the data used to train the model.

But how different would a model be for the February 2020 data? We first manipulated the data so that it was in the same grid as the September 2021 data and then performed some exploratory analyses. We found that again there was spatial correlation in the residuals between the areal units. We then followed the same procedures described above and fit the exact same hierarchical model using the February 2020 data. Summaries of the posterior distributions for the February 2020 data can be seen in Table 4.

If we compare the posterior distributions in Table 4 to the original posterior distributions in Table 3 we see that overall the values are quite similar. In many cases, they are almost exactly the same. The main differences are in the intercept coefficient and the presence of coral coefficient. This is interesting because the presence of coral data was the same used between the two models (since coral is a physical feature and does not change that quickly). We also note that when we predicted the log(Chlorophyll Concentration) of September 2021 using our model trained on February 2020 data we got an MSE of 0.6905, which is very similar to the reverse prediction noted above.

Undoubtedly, there are further improvements that could be made to this model. The first most obvious one is to include some sort of a temporal aspect. Training on data for each month across several years would allow

Table 4: (February 2020 Data) Results and summaries of the posterior distribution for all parameters from the model fit to the February 2020 data.

	Median	2.5%	97.5%	Effective Sample Size	Geweke Diagnostic
Intercept	1.2353	0.6542	1.8481	44.1	-1.9
Depth	-0.0001	-0.0002	-0.0001	35.7	-0.4
Salinity	-0.0391	-0.0571	-0.0221	43.6	1.8
Temperature	-0.0659	-0.0695	-0.0617	8.8	0.1
Coral Presence	0.1248	0.0873	0.1600	131.1	0.6
σ^2	0.0007	0.0005	0.0011	188.2	-0.2
τ^2	0.4990	0.4820	0.5166	5000.0	-2.6
ρ	1.0000	1.0000	1.0000	NA	NA

for seasonal effects to be captured. These would likely have a large impact given that chlorophyll production is a function of the amount of sunlight reaching the surface of the planet. Accounting for temporal effects might lead to a more robust model. Furthermore, additional data could be collected, such as cloud cover patterns or a measure of sunlight reaching the surface in order to further refine the model. Finally, this model could also be extended to other areas of the globe in order to carry out these same analyses in other parts of the world.

Conclusion

Although there are several limitations to the model, we found that it accurately predicts the log(Chlorophyll Concentration) and is a great tool for identifying phytoplankton "hot-spots", or areal units that are more likely to have phytoplankton colonies in the Indian Ocean. We found that there is spatial correlation in chlorophyll concentration even after accounting for factors such as temperature, salinity, depth, and the presence of coral. This supports the notion that biological life tends to clump together, and sometimes occurs at random given that the conditions are good. Furthermore, it provides researchers with target areas for scientific expeditions in order to study the phytoplankton's role in the ecology of our planet.

References

Field, C.B., Behrenfeld, M.J., Randerson, J.T., Falkowski, P., 1998. Primary Production of the Biosphere: Integrating Terrestrial and Oceanic Components. Science 281, 237–240. URL: https://www.science.org/doi/10.1126/science.281.5374.237, doi:10.1126/science.281.5374.237.

Graham, S., Taylor, J., Joyner, E., 2022. My NASA Data. URL: https://mynasadata.larc.nasa.gov/.

Lee, D., 2013. CARBayes: An R package for Bayesian spatial modeling with conditional autoregressive priors. Journal of Statistical Software 55, 1–24. URL: https://www.jstatsoft.org/htaccess.php?volume=55&type=i&issue=13.

Li, G., Lin, Q., Ni, G., Shen, P., Fan, Y., Huang, L., Tan, Y., 2012. Vertical Patterns of Early Summer Chlorophyll a Concentration in the Indian Ocean with Special Reference to the Variation of Deep Chlorophyll Maximum. Journal of Marine Biology 2012, 1–6. URL: http://www.hindawi.com/journals/jmb/2012/801248/, doi:10.1155/2012/801248.

Pante, E., Simon-Bouhet, B., 2013. marmap: A package for importing, plotting and analyzing bathymetric and topographic data in r. PLoS ONE 8, e73051. Doi:10.1371/journal.pone.0073051.

Wu, H., Zou, D., Gao, K., 2008. Impacts of increased atmospheric CO2 concentration on photosynthesis and growth of micro- and macro-algae. Science in China Series C: Life Sciences 51, 1144–1150. URL: http://link.springer.com/10.1007/s11427-008-0142-5, doi:10.1007/s11427-008-0142-5.